



SAVEETHA SCHOOL OF ENGINEERING



SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA51 – CRYPTOGRAPHY AND NETWORK SECURITY

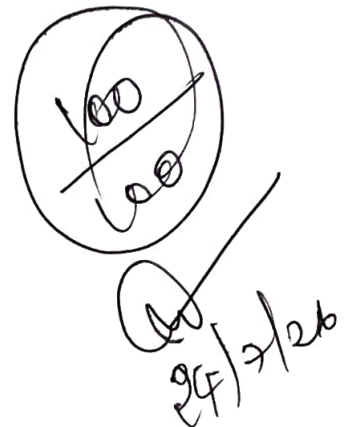
Assignment1

Question1:

- **Encrypt** the message "DEFEND THE WALL" using Caesar cipher with key = 3.
- **Decrypt** the message "KHOOR ZRUOG" using Caesar cipher with key = 3.
- A Caesar cipher is used to encrypt the word "SECRET" to "VHFUHW".
Determine the key used, and decrypt "WKLV LV D WHVW".

Question2.

- Using the substitution key:
A→Q, B→W, C→E, D→R, E→T, F→Y, ..., Z→M
Encrypt the word "FIRE".
- Given this cipher text "UJHHD QEB NRFZH YOLTK",
and knowing it is encrypted with a **monoalphabetic substitution**,
Find the plaintext using frequency analysis (assume it's English).



Question3.

- **Encrypt** the message "ATTACKATPALACE" using the keyword "LEMON".
- **Decrypt** the cipher text "LXFOPVEFRNHR" using the keyword "LEMON".
- Given the encrypted message "QNXEPKMAEGTU" using the Vigenère cipher and keyword "KEY",
Find the original message.

Question4.

Given a 2x2 key matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

- **Encrypt** the message "HI" using the Hill cipher.
(Use A=0, B=1, ..., Z=25.)
- If the message "PO" was encrypted to "QC" using the above Hill cipher matrix,
Verify the result by showing the matrix multiplication.

Question5.

- **Encrypt** the message "MEETMEATERLUNCH" using the keyword "ZEBRAS".
- **Decrypt** the message "EANMTEFCUHETREL" using the keyword "BOARD".

Question6.

- Using the one-time pad key **XMCKL**, encrypt the message **"HELLO"**.
- Explain why the message **"ATTACKNOW"** encrypted using OTP with key **"QWERTYUIO"** is **unbreakable** under ideal conditions.

Question7.

- **Encrypt the message:**
 - ♦ Plaintext: **"WE ARE DISCOVERED"**
 - ♦ Keyword: **"ZEBRA"**

Steps expected:

- Arrange the message in rows under the keyword.
- Determine column order based on alphabetical sorting of keyword letters.
- Read columns in sorted order to form ciphertext.

- **Decrypt the message:**
 - ♦ Ciphertext: **"EVLNEACDTKESEAQROFOJDEECUXWN"**
 - ♦ Keyword: **"TRIAL"**

Task:

- Reconstruct the matrix based on the keyword.
 - Rearrange the columns to the original order.
 - Read row-wise to get the plaintext.
- The message **"INFORMATIONSECURITY"** is encrypted using columnar transposition with keyword **"HACK"**.
 1. Encrypt the message.
 2. Explain how many rows are needed.
 3. Add necessary padding characters (like 'X') if needed.

Question8.

You are given the following cipher text encrypted using a **monoalphabetic substitution cipher**:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUZUETUCG

Assume it is an English sentence.

Task:

1. Perform **frequency analysis** of letters.
2. Use **standard English letter frequencies** (e.g., E, T, A, O are most common) to **hypothesize probable mappings**.

3. Attempt to **partially decrypt** the message.
4. Provide the best guess for the plaintext message (even partial guesses allowed).

Question9.

You intercepted a message encrypted with a monoalphabetic substitution cipher:

Ciphertext: BJJW QJW YMJW NX F XNRUQJWX

You know the original plaintext starts with:
"THIS IS THE".

Task:

1. Use the known-plaintext portion to **find substitution mappings**.
2. **Complete the decryption** of the full message.
3. Validate if any letters repeat and adjust your substitution table if necessary.

Question10.

You are given the cipher text below, encrypted using the Vigenère cipher with an unknown short key (2-5 letters long):

Ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Try to recover the key and decrypt the full message.

Question 1 :-

- Encrypt the message "DEFEND THE WALL" using Caesar cipher with Key = 3

Ans :- Given plain text :- DEFEND THE WALL

Key = 3

Cipher text = GHIHQH WKH ZDQD

- Decrypt the message "KHOOR ZRUOG" using Caesar cipher with Key = 3

Ans :- Given, cipher text = KHOOR ZRUOG

Key = 3.

plain text = HELLO WORLD

- A Caesar cipher is used to encrypt the word "SECRET" to "VHFUHW". Determine the key used, and decrypt "WKLV LV D WHUV".

Ans :- Shift $\Rightarrow S \rightarrow V = +3$

Decrypt : WKLV LV D WHUV

The plain text : This is A TEST

Question :- 2

• using the substitution key :

$A \rightarrow Q, B \rightarrow W, C \rightarrow E, D \rightarrow R, E \rightarrow T, F \rightarrow Y \dots Z \rightarrow M$

encrypt the word "FIRE".

Ans :- Key : $A \rightarrow Q, B \rightarrow W, C \rightarrow E, D \rightarrow R, E \rightarrow T, F \rightarrow Y$

Encryption :- $F \rightarrow Y$

$I \rightarrow O$

$R \rightarrow K$

$E \rightarrow T$

$\therefore$ The plain text is YOKT

• Given this cipher text "UJHHD QEB NRFZH YOLTK" and knowing it is encrypted with a monoalphabetic substitution find P.T

Ans :-

U	J	H	H	D	Q	E	B	N	R	F	Z	Y	O	L	T	K
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The letter H appears most, so it likely to represent one of the common english letter as E or T

UJHHD
THE

QEB
QUICK

NRFZH
BROWN

YOLTK
FOX

$\therefore$ The plain text :- THE QUICK BROWN FOX.

Question :- 3

- encrypt the message "ATTACK AT PALACE" using the keyword "LENON":

Decryption calculation

Cipher	Key	$(C - K + 26) \bmod 26$	plain
16	10	6	G
13	4	9	J
23	24	25	Z
4	10	20	U
15	4	11	L
10	24	12	M
12	10	2	C
0	4	22	N
4	24	6	S
6	10	22	N
19	4	15	P
20	24	22	N

Plain text : GJZULMCWGNPW

Question:-4

Given a 2×2 Key matrix :

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

• Encryption formula is $C = K \times P \pmod{26}$

• Encrypt the message "HI" using the cipher matrix

letter : H I

value : 7 8

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$= (3 \times 7) + (3 \times 8) = 21 + 24 = 45$$

$$= (2 \times 7) + (5 \times 8) = 14 + 20 = 34$$

$$C = \begin{bmatrix} 45 \\ 34 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 \\ 8 \end{bmatrix} = \begin{bmatrix} T \\ C \end{bmatrix}$$

$\therefore$ Cipher text = TC

• verify that 'PO' is encrypted to 'TC' using the same Hill cipher matrix.

Ans: letter P O
value 15 14

$$P = \begin{bmatrix} 15 \\ 14 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 15 \\ 14 \end{bmatrix}$$

$$(3 \times 15) + (3 \times 14) = 45 + 42 = 87$$

$$(2 \times 15) + (5 \times 14) = 30 + 70 = 100$$

$$C = \begin{bmatrix} 87 \\ 100 \end{bmatrix} \Rightarrow \text{mod } 26 = \begin{bmatrix} 9 \\ 22 \end{bmatrix} \Rightarrow \begin{bmatrix} J \\ W \end{bmatrix}$$

$\therefore PO \rightarrow JW$

Question 5:

• Encrypt the message "MEETMEATERLUNCH" using "ZEBRAS".

Ans: Keyword = "ZEBRAS"
↓ ↓ ↓ ↓ ↓ ↓ ↓
6 3 2 4 1 5

Z	E	B	R	A	S
M	E	E	T	M	E
A	F	T	E	R	L
U	N	C	H	X	X

- Decrypt the ciphertext "LXFOPVEFRNHR" using "LEMON":

Ans:-

Cipher text:	L X F O P V E F R N H R
Keyword:	L E M O N L E M O N L E

Decryption calculation.

Cipher	Key	$(C - K + 26) \bmod 26$	Plain Text
11	11	0	A
23	4	19	T
5	12	19	T
14	14	0	A
15	13	2	C
21	11	10	K
4	4	0	A
5	12	19	T
17	14	3	D
13	13	0	A
7	11	22	W
17	4	13	N

plain text: ATTACK AT DAWN.

- Decrypt the ciphertext "QNXEPKHAEGTU" using "KEY"

Ans:-

Ciphertext	Q N X E P K M A E G T U
Keyword	K E Y K E Y K E Y K E Y

plain text = A T T A C K A T P A L A C E

Key word = L E M O N L E M O N E E M O

convert letters into numbers

letters :- A T T A C K A T P A L A C E

value :- 0 19 19 0 2 10 0 19 15 0 11 0 2 4

Keyword :-

letters :- L E M O N L E M O N L E M O

value :- 11 4 12 14 13 11 4 12 14 13 11 4 12 14

Encryption calculation :-

Plain	Key	$(P+K) \bmod 26$	Cipher
0	11	11	L
19	4	23	X
19	12	5	F
0	14	14	O
2	13	15	P
10	11	21	V
0	4	4	E
19	12	5	F
15	14	3	D
0	13	13	N
11	11	22	S
0	4	4	E
2	12	14	O
4	14	18	S

Cipher text :- LXFOPEFDNNEOS.

1(A column) $\rightarrow$ MRX $\rightarrow$ MRX

2(B column) $\rightarrow$ ET

3(E column) $\rightarrow$ EFN

4(R column) $\rightarrow$ TEH

5(S column) $\rightarrow$ ELX

6(Z column) $\rightarrow$ HAU

Cipher text = MRXETCEFNTEHELEXHAU

- Decrypt the ciphertext "EANNHTECTUHTREL" using "BOARD"

Ans:

Keyword : B O A R D
 $\downarrow \downarrow \downarrow \downarrow \downarrow$
2 4 1 5 3

Total letters = 15

columns = 5

Rows = $15 \div 3 = 3$

create 3x5 matrix

B	O	A	R	D
M	H	E	R	F
T	E	A	E	C
E	T	N	L	U

Plain text : MHERFTEAECTNLU

Question 6:

- using the one-time pad key XMCKL, encrypt the message "HELLO".

Ans:

P.T = H E L L O
7 4 11 11 14

X M C K L

23 12 2 10 11

$$C = (P + K) \bmod 26$$

Plain(P)	Key (K)	P+K	Mod 26	Cipher
7	23	30	4	E
4	12	16	16	Q
11	2	13	13	N
11	10	21	21	V
14	11	25	25	Z

Ciphertext : EQNVZ

HELLO + XNCKL = EQNVZ

- explain why the message "ATTACK NOW" encrypted using OTP with Key "QWERTYUIO" is unbreakable.

- Ans:-
1. The Key is truly random
 - Every letter in the key is generated randomly
 2. The Key length is equal to the plaintext length
 - Here, the plaintext "ATTACK NOW" has 9 letters
 - The Key "QWERTYUIO" also has 9 letters
 3. The Key is used only once.
 - Reusing the same key makes OTP vulnerable to attacks
 4. The Key is kept completely secret
 - only the sender and receiver know the key

Question : 7

(a) Encrypt the message.

P.T :- WE are discovered

Keyword : ZEBRA
5 3 2 4 1

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

Cipher text : EODASREIERCEWDV

(b) Decrypt the ciphertext.

C.T :- EVLNEACDTKESAO ROFOJDEECUXWN

Keyword : TRIAL
5 4 2 1 3

$$C.T :- 28 \div 5 \Rightarrow [28 \div 5] = 6$$

T	R	I	A	L
E	V	L	N	E
A	C	D	T	K
E	S	A	O	R
O	F	O	D	E
E	C	U	X	W
N				

Plain text : WE are discovered flat once.

(c) encrypt "information security" using keyword Hack

HACK
3 1 2 4

length = 19, columns = 4

$$[19/4] = 5$$

H	A	C	K
I	N	F	O
R	M	A	T
I	O	N	S
E	C	U	R
I	T	Y	X

Cipher text : NMOC TFANUYIRE IOTSRX

Question :- 08

Cipher text encrypted using a monoalphabetic substitution cipher "UZQ SOVUOHTXNOPVG POZPEVSGZWSZWSZO PFPSXUZUE TO CH" Assume it is English

1. count the frequency of each letter

letter : Z O U P S E V G X T Q H M W C F

frequency : 8 6 5 5 5 4 3 3 2 2 1 1 1 2 1 1

Z → 8 times, O → 6 times, U, P, S → 5 times

2. compare with standard English letter frequencies.

E → Highest, T → second, A → third, O → fourth, I → fifth, N → sixth

Cipher letter : Z O O P S E V G

Possible plain letter : E T A O N I R S

Cipher :	U Z Q S O V H X M P G W E T C F
Plain (Guess) :	A E ? N T R ? ? ? O S ? I ? ? ?

* Using only frequency analysis, the plaintext cannot be completely recovered. The above decryption is the best possible estimate without additional clues.

Question :- 09 :-

Encrypted with a monoalphabetic substitution cipher

C.P: BJJW QJN YMJN NXF XNRUQJWX, P.T is
THIS is THE .

Ans :- C.T $\Rightarrow$ BJJW QJN YMJN NXF XNRUQJWX

P.T $\Rightarrow$ THIS is THE .

Cipher :-	B J J W Q J N Y M J N
Plain :-	T H I S I S T T H E ?

Cipher	Plain
BJJW	THIS
QJN	IS
YMJN	THE
NX	AS
F	A
XNRUQJNX	SIMPLEST

THIS IS THE A SIMPLE TEST.

- J appears many times and consistently map to H.
- u appears repeatedly and map to I.

Question :- 10

You are given the cipher text below, encrypted using the Vigenere cipher with an unknown short key (2-5 letter long) :-

C.T $\Rightarrow$ ZICVTWQNGRZGVTVNAVZHCOYGLMHJ.

Ans :- Key length is 2-5 letters

Assume the Key = 3:

Position:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
Cipher:	Z	I	C	V	T	W	Q	N	G	R	Z	G	V	T	N	A	V	Z	H	C	O	Y	G	L	M	H	J

Group 1:- ZVQRVAHYM

Group 2:- ITNZTVCHG

Group 3:- CWGHWZQLT

KEY: K $\rightarrow$ 10, E $\rightarrow$ 4, Y $\rightarrow$ 24

$$P = (C - K + 26) \bmod 26$$

Cipher	Z	I	C	V	T	W	Q	N	G	R	Z	G	V	T	N	A	V	Z	H	C	O	Y	G	L	M	H	J
Key	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	Y	K	E	Y
Plain	P	E	E	L	P	Y	G	J	I	H	V	I	L	P	Y	Q	R	B	X	Y	S	O	C	N	C	C	L

Plain text :- PEELPYGJIHVILPYQRBXYSOCNCCL